



## Feathers

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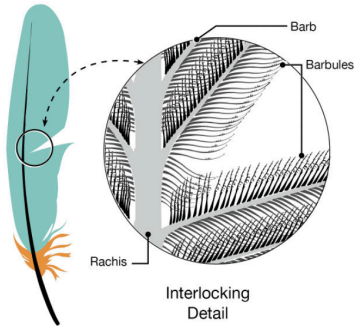
# Feathers!

- ▶ An extant avian apomorphy (shared with some theropods)
- ▶ Mostly made from beta-keratin—strong protein polymer unique to birds and reptiles

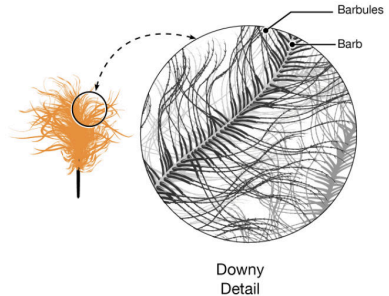


# Feather anatomy

- ▶ An extant avian apomorphy (shared with some theropods)
- ▶ Mostly made from beta-keratin—strong protein polymer unique to birds and reptiles



Pennaceous



Plumulaceous

# Feather types

- ▶ Wing (remige) and tail (retrix) help with flight, mobility, displays
- ▶ Contours cover body
- ▶ Down and semiplumes insulate, lack hooks on barbules; differ by presence of rachis in latter
- ▶ Bristles and filoplumes have sensory purposes



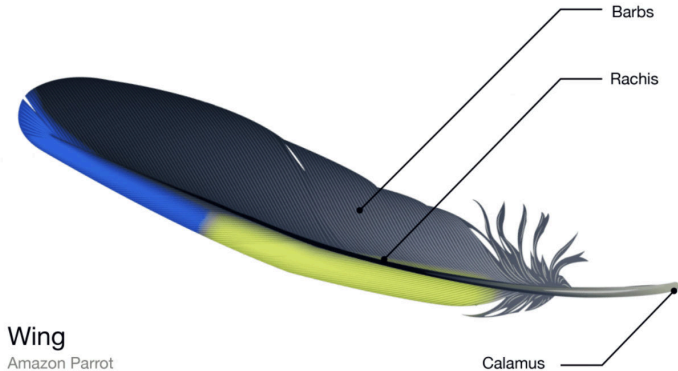
## Powder down

- ▶ Found in herons, pigeons, doves, tinamous, bustards and some parrots
- ▶ Down feathers that never molt, grow continuously, and break down into a talcum-powder like substance at the tip



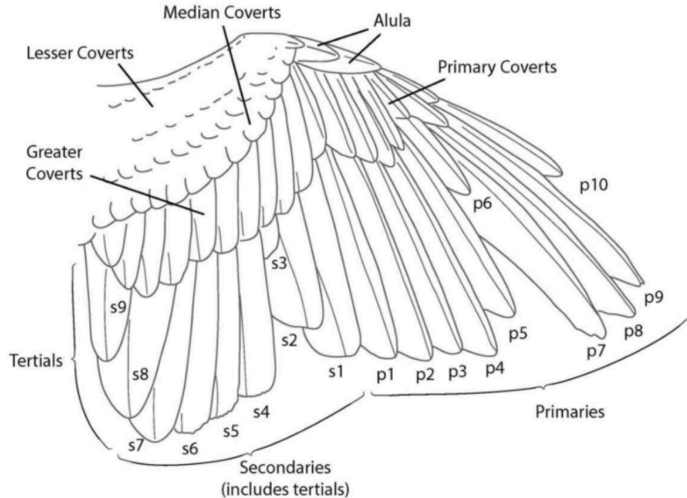
# Remiges

- ▶ Asymmetrical for flight
- ▶ Barbs have hooked barbules of central rami (ramus)
- ▶ Planar surface on either side of rachis called “vanes”



# Remiges

- ▶ 9-12 primaries (attach to carpometacarpus)
- ▶ 6-40 secondaries (attach to ulna)



# Remiges

- ▶ 6 secondaries in hummingbirds
- ▶ 40 secondaries in albatross



(UPS Wing and Tail Image Collection)

# Modified remiges



<https://www.youtube.com/watch?v=tSHjhCN6NC0>

(*Caprimulgus longipennis*)

## Modified remiges

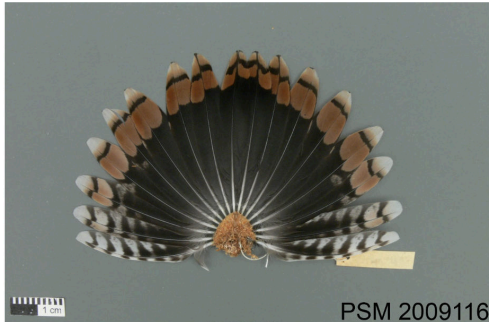
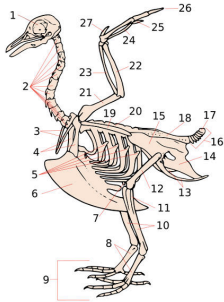


<https://macaulaylibrary.org/asset/253493131>

(*Machaeropterus deliciosus*; Cornell)

# Retrices

- ▶ Usually 12 (24 in snipe)
- ▶ attach to pygostyle



<https://macaulaylibrary.org/asset/94350>

(UPS Wing and Tail Image Collection)

## Modified retrices

- ▶ Sexual selection can modify retrices for display (birds of paradise)
- ▶ Natural selection can modify retrices for foraging niche (woodpeckers)

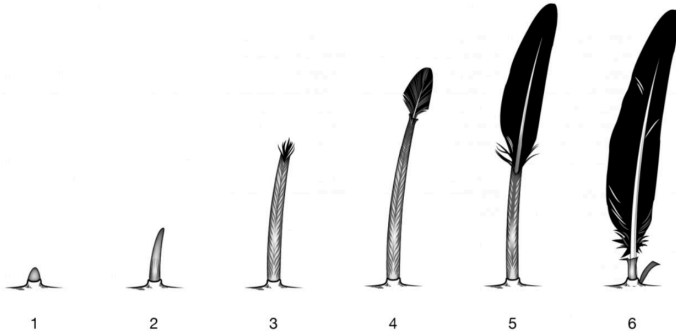


<https://www.youtube.com/watch?v=fHDHJE15NKk>

(*Astrapia mayeri*)

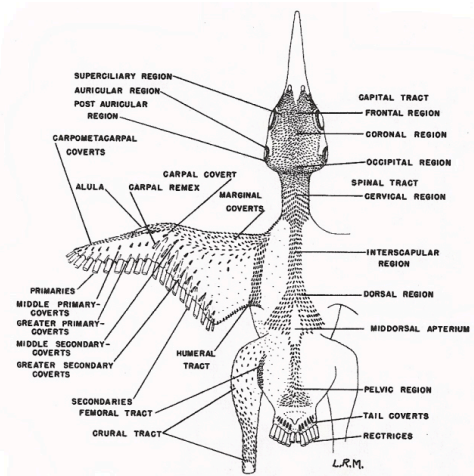
# Feather development: Stages

- ▶ Epidermis thickens to **placaode**, elongates into **feather germ**, invaginates to form **follice**, develops barbs and rachis inside **sheath** that sheds when growth is complete
- ▶ Complete feathers are dead structures!



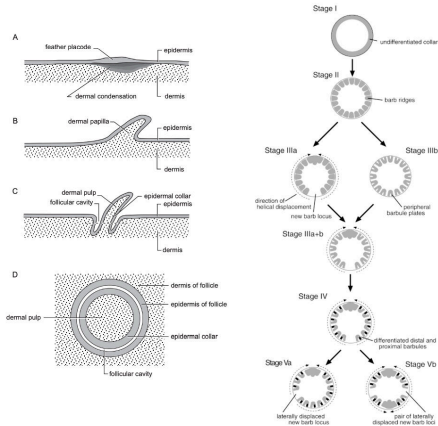
# Feather tracts

- ▶ Feather tracts are called pterylae
- ▶ Regions between tracts called apteria

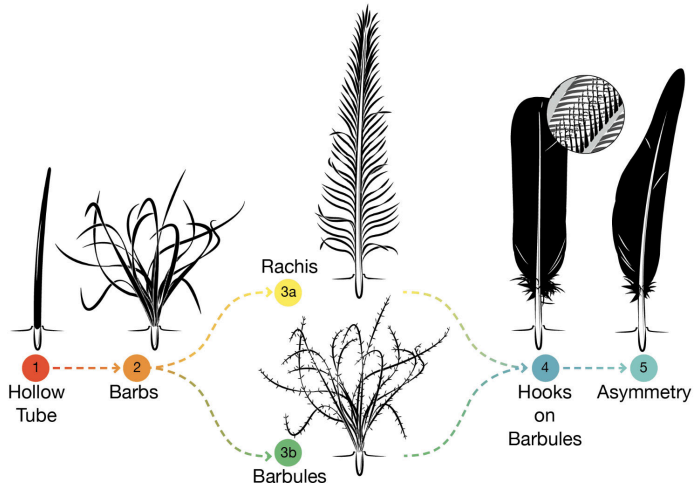


# Evo-devo feather development hypothesis

- ▶ Feather morphology evolution inferred by order of dependency of features during development (e.g., rachis is formed by fusion of barbs)

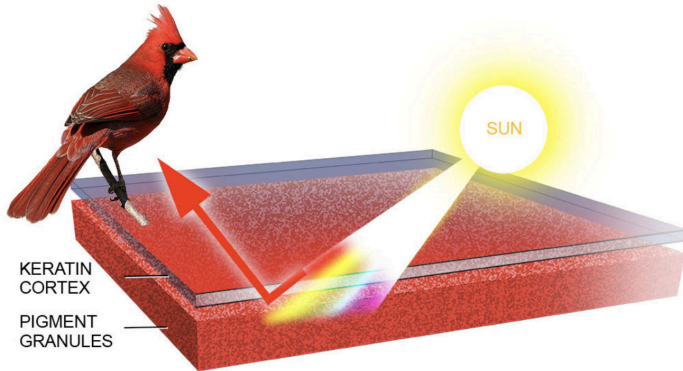


# Feather evolution summarized



(Cornell)

# Feather coloration: pigments



(Cornell)

# Melanins

- ▶ Pigments responsible for browns and blacks
- ▶ Produced by melanosome organelles; transferred to feathers and retained in beta-keratin
- ▶ Makes feathers more durable and resistant to bacteria



(*Bubo virginianus*)

# Carotenoids

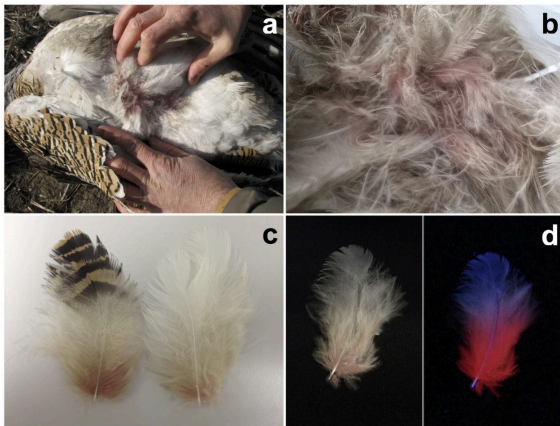
- ▶ Pigments responsible for reds, yellows, and oranges
- ▶ Obtained from dietary sources
- ▶ Can interact with melanins to create olive-green



(*Setophaga fusca*)

# Porphyrins

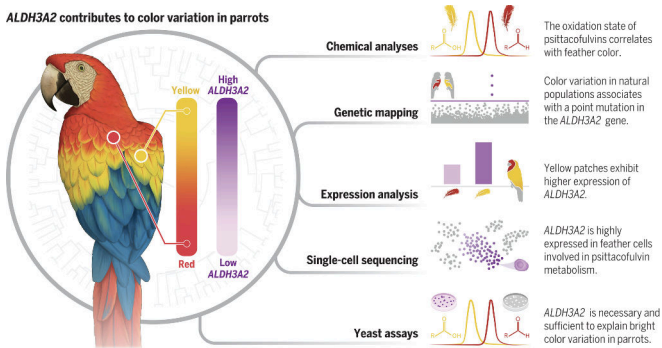
- ▶ Chemically similar to chlorophyll and heme (in hemoglobin)
- ▶ Browns and reds; degrade easily under light
- ▶ Creates short-lived color that may signal virginity in bustards!



# Psittacofulvins

- ▶ Pigments responsible for orange, yellow, and bright red in parrots
- ▶ Synthesized in feather cells; color variation from mutation impacting oxidation state

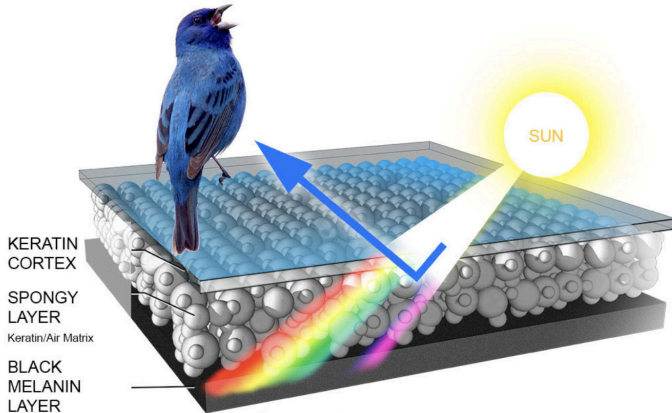
*ALDH3A2* contributes to color variation in parrots



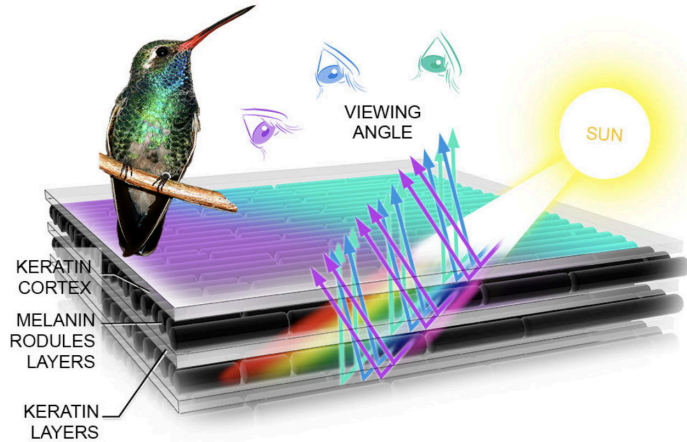
(Arbore et al. 2024)

# Feather coloration: structural color

- ▶ Blues; can combine with carotenoids to produce greens (except in Turacos; “turacoverdin”)



# Feather coloration: iridescence



(Cornell)

What pigments are likely responsible for the *remiges* of this Golden parakeet?



(*Guaruba guarouba*)

- a) psittacofulvins
- b) carotenoids
- c) structural color
- d) structural color + carotenoids
- e) none of the above

# Feather care

- ▶ Preening with bill and feet: arrange barbs, remove parasites
- ▶ Waterproof with secretions from uropygial gland



(Wikipedia)

# Feather replacement: molt

- Birds replace old feather 2-3x during their first year; 1-2x / year after

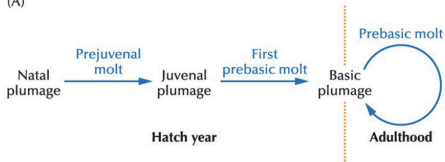


(IBP)

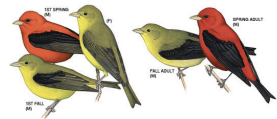
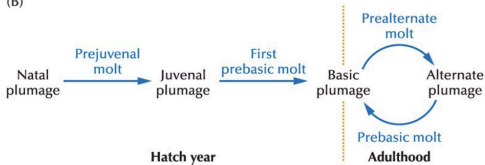
# Classic molt patterns

- ▶ One molt per year: chickadees, hawks, hummingbirds, jays, etc
- ▶ One complete / one partial molt per year: tanagers, warblers, buntings

(A)



(B)



This Orange-crowned Warbler bred in Glacier NP and is about to migrate to Mexico. What molt did it last undergo?

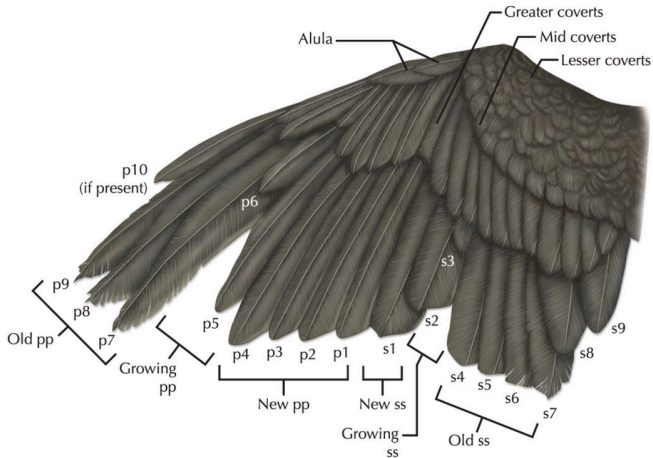


(*Leiothlypis celata*)

- a) prealternate molt
- b) prebasic molt
- c) prejuvenal molt
- d) first prebasic molt

# Centrifugal molt

- ▶ Many species (esp. passerines) begin with first primary and molt outwards towards p10



# Synchronous molt

- ▶ Waterfowl molt all primaries simultaneously, lose ability to fly for several weeks
- ▶ Males in basic plumage during this stage



(Marlin Harms)

## “Molt” by wear

- ▶ Plumage can also change by wear and abrasion, e.g. Snow Buntings



(Birds of the World)

## References

- ▶ Title Slide Species: *Pelecanus erythrorhynchos*, *Sheppardia sharpei*, *Gyps fulvus*, *Oriolus chlorocephalus*
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## References

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